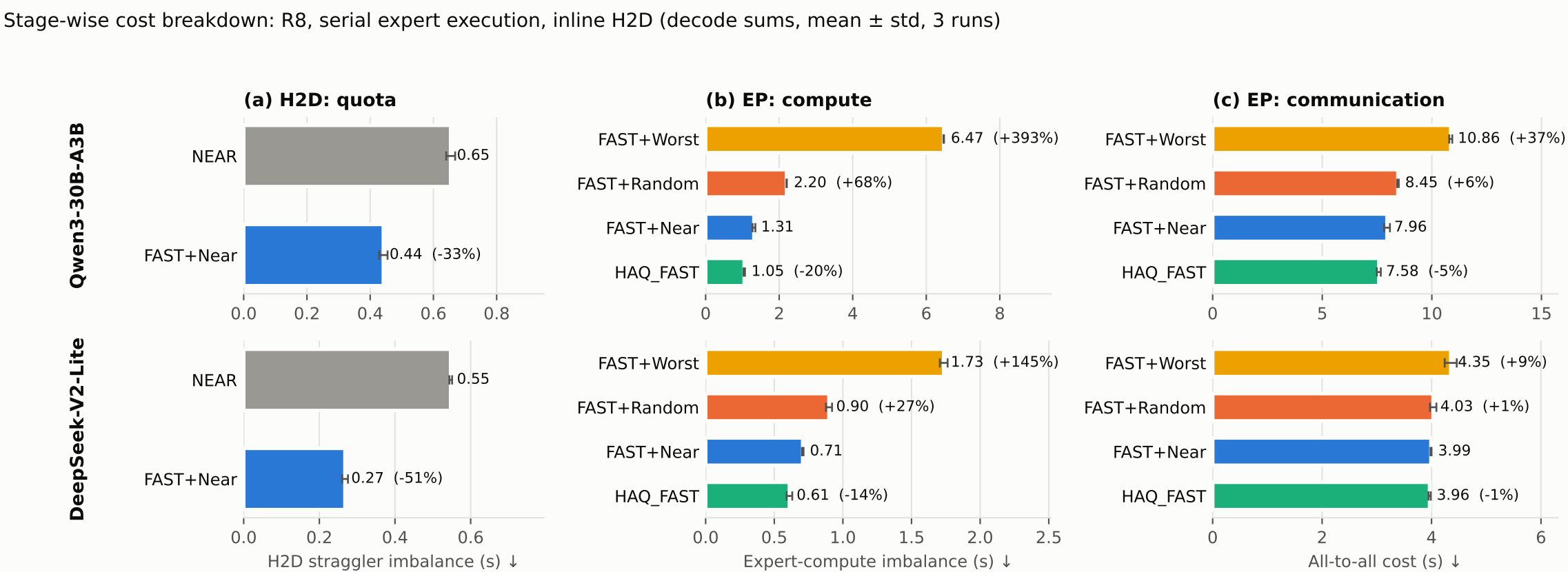
Stage-wise cost breakdown: R8, serial expert execution, inline H2D (decode sums, mean ± std, 3 runs)
(a) H2D: quota
(b) EP: compute
(c) EP: communication
Qwen3-30B-A3B
DeepSeek-V2-Lite
NEAR
FAST+Near
FAST+Worst
FAST+Random
HAQ_FAST
0.65
0.44 (-33%)
0.55
0.27 (-51%)
6.47 (+393%)
2.20 (+68%)
1.31
1.05 (-20%)
1.73 (+145%)
0.90 (+27%)
0.71
0.61 (-14%)
10.86 (+37%)
8.45 (+6%)
7.96
7.58 (-5%)
4.35 (+9%)
4.03 (+1%)
3.99
3.96 (-1%)
H2D straggler imbalance (s) ↓
Expert-compute imbalance (s) ↓
All-to-all cost (s) ↓
(a) relative to NEAR; (b)(c) relative to FAST+Near. HAQ_FAST uses a hit-aware quota (H2D imbalance stays near FAST: Qwen 0.46 vs 0.44 s, DeepSeek 0.32 vs 0.27 s).